

SMART IRRIGATION ROVER

ESP32-Based Adaptive Dry-Zone Irrigation System

A low-cost autonomous rover that detects dry soil areas, records repeated dry zones through ESP32 logic, and waters only the required spots.

Prototype Innovation

No SD Card Needed

Bangla AI Support

PROJECT INNOVATION

The Smart Irrigation Rover introduces ESP32-based adaptive dry-zone irrigation. Unlike a basic irrigation robot that only reacts to current moisture, this rover uses lightweight dry-zone memory logic to identify repeatedly dry areas and prioritize them for watering.

The innovation is not external storage hardware. It is the decision algorithm: detect dry soil, mark the zone, compare repeated dryness, and water only where needed.

CORE INNOVATION FEATURES

- **ESP32 Dry-Zone Memory**
Records small dry-zone points using ESP32 logic and identifies repeated dry areas.
- **Location-Aware Spot Watering**
Waters only the specific dry zones instead of watering the entire field.
- **Adaptive Priority Decision**
Prioritizes watering using current soil moisture and previous dry-zone history.
- **Real-Time Sensor Monitoring**
Reads soil moisture and rover status for field condition monitoring.
- **Low-Cost Practical Design**
Uses affordable components for small farms, gardens, and course demonstrations.
- **Bangla Web AI Support**
A website AI assistant explains the project through Bangla voice interaction.

PROTOTYPE ACCURACY NOTE

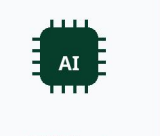
In this prototype, field location is represented as simple zones or rover movement steps. The ESP32 can store and compare small records in internal logic. Future versions can add GPS, wheel encoder, or cloud database for more accurate field mapping.

HOW IT WORKS - STEP BY STEP

- 1

**Move & Scan**
Rover reads soil moisture while moving.
- 2

**Detect Dry Zone**
Low moisture marks the area as dry.
- 3

**ESP32 Logic**
Controller checks moisture and dry-zone history.
- 4

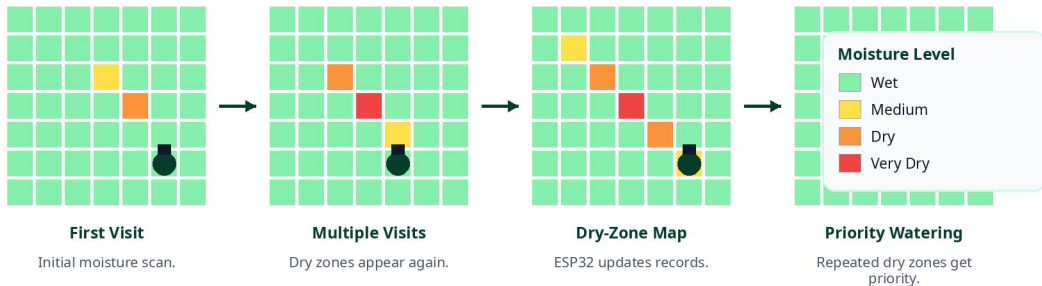
**Priority Decision**
Repeated dry zones receive higher priority.
- 5

**Spot Watering**
Pump waters only the required spot.
- 6

**Status Update**
Records and dashboard status are updated.

DRY-ZONE MEMORY LOGIC

The rover divides the field into simple zones. Each zone can be Wet, Medium, Dry, or Very Dry. Repeated dry zones are prioritized for the next watering cycle. No SD card module is required for this prototype.



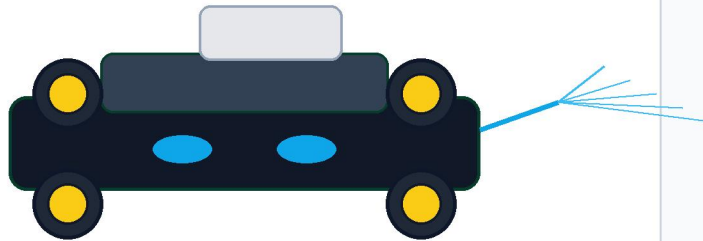
Key point: The innovation is the adaptive dry-zone decision algorithm - not a memory card. The ESP32 keeps lightweight zone records and uses them for smarter watering decisions.

SYSTEM FLOW



This flow enables the rover to make watering decisions from sensor data, while keeping the system low-cost and suitable for a prototype demonstration.

ROVER DESIGN - HARDWARE OVERVIEW



-  ESP32 Controller
-  Soil Moisture Sensor
-  Ultrasonic Sensor
-  4WD Motors
-  Motor Driver
-  Water Pump
-  Spray Nozzle
-  Battery Supply

EXAMPLE SCENARIO

- 1

Rover reaches Zone A
- 2

Moisture is low
- 3

Zone A is marked dry
- 4

Rover waters Zone A
- 5

Next visit checks Zone A
- 6

Repeated dryness gets priority

EXPECTED BENEFITS

-  Reduces unnecessary watering
-  Saves water and battery energy
-  Supports precise irrigation
-  Improves decisions over visits
-  Works with low-cost components
-  Suitable for small farms and gardens

FUTURE IMPROVEMENTS

GPS or wheel encoder for accurate field mapping
Cloud database for long-term history
Automatic route planning
AI-based irrigation prediction
Voice commands connected with ESP32